🚰 Plant Plumbing

Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 4–6 hours

💧 Colour Climb: Watch the Xylem

You will learn: Xylem carries water up the stem; vascular bundles are in a ring in dicots and scattered in monocots.

YOU NEED

- A balsam (or celery) stem with leaves
- A maize or grass stem
- Two glasses of water
- Red or blue food colour (or ink)
- A blade (with an adult)

⚠️ **Safety:** Use the blade only with an adult. Food colour may stain clothes.

STEPS

1. Add food colour to both glasses of water.
2. Cut the stems at the base and stand one in each glass.
3. Note the time; check every hour where the colour has reached.
4. After 4–6 hours, cut a thin cross-section near the top of each stem.
5. Look at the section with the phone microscope and sketch the coloured dots.
6. Compare the pattern in balsam and maize.

OBSERVATION TABLE

Time (h)	Balsam: colour reached	Maize: colour reached

QUESTIONS

1. Which tissue carried the colour?

2. How was the pattern different in maize and balsam?

3. Would the colour rise faster on a hot, windy day? Why?

WHAT IS HAPPENING? — THE SCIENCE

The coloured dots are xylem in vascular bundles. In a dicot stem (balsam) they form a ring; in a monocot stem (maize) they are scattered. Water rises through xylem vessels pulled by transpiration from the leaves.

Key terms: **xylem** **vascular bundle** **dicot stem** **monocot stem**

Activity 2 · Hands-on · 🧪 In the lab · ⏱ 30 minutes

🍃 Nail-Polish Stomata Count

> **You will learn:** Stomata are guarded by bean-shaped guard cells (dumb-bell shaped in grasses); most are on the lower surface of dicot leaves.

YOU NEED

- A dicot leaf (hibiscus, rose) and a grass leaf
- Clear nail polish
- Clear tape
- Glass slide
- Phone microscope

> ⚠️ **Safety:** Use nail polish in a well-ventilated room and keep it away from flames.

STEPS

1. Paint a thin patch of nail polish on the upper and lower surfaces of the dicot leaf, and on the grass leaf.
2. Let it dry for 10 minutes.
3. Press tape on the patch, peel it off and stick it on the slide.
4. Observe at high zoom. Count stomata in one field of view.
5. Repeat for the other surfaces and sketch the guard cells.

OBSERVATION TABLE

Leaf surface	Stomata in field 1	Field 2	Average	Guard cell shape

QUESTIONS

1. Which surface had more stomata? Why is this useful?

2. Draw a dumb-bell shaped guard cell.

3. What are subsidiary cells?

WHAT IS HAPPENING? — THE SCIENCE

The polish makes an impression of the epidermis. Dicot leaves usually have more stomata on the lower surface, which reduces water loss. Grass guard cells are dumb-bell shaped; dicot guard cells are bean-shaped.

Key terms: **stomata** **guard cells** **epidermis** **subsidiary cells**

Digital activity

💻 Tissue Sorter

> Tap a feature, then tap the tissue it describes.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu1-ch6/2**

PARENCHYMA · COLLENCHYMA · SCLERENCHYMA · XYLEM · PHLOEM

Parenchyma	Collenchyma	Sclerenchyma	Xylem	Phloem

Thin walls, stores food · Chlorenchyma in leaves · Corners thickened with pectin · Supports young stems and petioles · Dead cells with lignin · Makes guava pulp gritty · Tracheids and vessels · Carries water upwards · Sieve tubes and companion cells · Carries food

> **You will learn:** Simple tissues (parenchyma, collenchyma, sclerenchyma) have one cell type; complex tissues (xylem, phloem) have several.

THINK & WRITE

1. Which tissue has living cells without a nucleus at maturity?

2. Which simple tissue is dead at maturity?